Game Design Document

By Liam Treacy (c00298242)

What is this game?

Hank's Odyssey is a 3D platformer being developed in Unity for my final year project. The game focuses on platforming, exploring environments, and defeating enemies across three distinct environments. Each stage will introduce different surroundings, enemies and potential combat options, with the difficulty increasing as the player progresses.

Features of Hank's Odyssey

Movement and Combat

Movement will be done for the time being WASD/Arrow keys. However, this is planned to only be a placeholder. I want to attempt to add controller support with an Xbox Series S controller. I watched some YouTube videos on how controller input works.

https://www.youtube.com/watch?v=p-3S73MaDP8



Movement will not just be about direction however, there will be other features on top of that. Jumping will be a feature, it is currently included with the space bar, however I would like to change this to the A button of the Xbox controller.

I would like to add some combat features, the idea is either punching and kicking, or picking up weapons and being able to shoot, being assigned to a certain key and a certain button the Xbox controller. The combat would will touched upon later in the enemies section.

https://www.youtube.com/watch?v=8kKLUsn7tcg



Since the game is set in 3 distinct areas, the weapons would need to reflect that, stage 1 would have stuff like rusty knives and trash can lids. Stage 2 is a more natural forest stage with branches and rocks. Stage 3 is in the caves so sharp, shiny crystals would be used to attack the cave monsters.

Another possible idea is being able to punch and kick, and once a weapon is equipped you lost the ability to punch, but you can now use the weapon, this is a feature I will discuss with my supervisor Muhammed.

Level Design

The game will be split into three stages, with each stage having its own environment and visual style. The levels will be designed around platforming, jumping, avoiding obstacles, and reaching the end platform.

The levels should be structured in a way that allows the player to understand where they need to go, while still providing challenges along the way. Platforms will be placed to test the player's movement and jumping skills, with enemies and other obstacles providing additional difficulty. Each level will be more difficult than the last

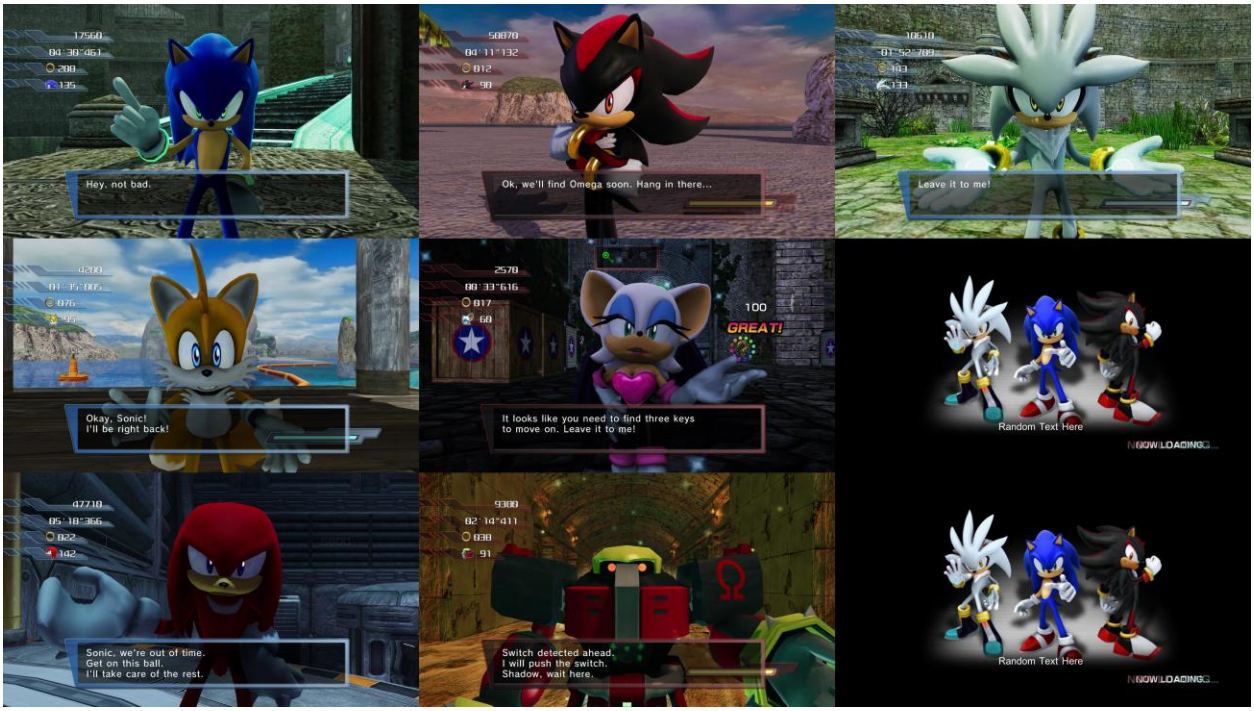
Each stage will have an end point to show the player that they have completed the level. I also plan to use visual differences between the environments to make progression through the game more noticeable. The stages should take roughly a few minutes to complete each but I am unsure of the length as of yet.

Enemies and Scoring

Enemies will provide an additional challenge during the platforming sections. Hank will be able to defeat enemies by jumping on them. When an enemy is defeated, it will be removed from the scene and treated as dead, the player will receive points. Different enemies can award different amounts of points depending on how difficult they are to defeat. Some enemies may need multiple hits and have a health bar, and some enemies may be immune to certain attacks. For example, a weaker enemy can give you 50 points than a more large and in charge enemy which will give 100 points. Sound effects will also be used when enemies are defeated to give the player feedback that the action has been successful.

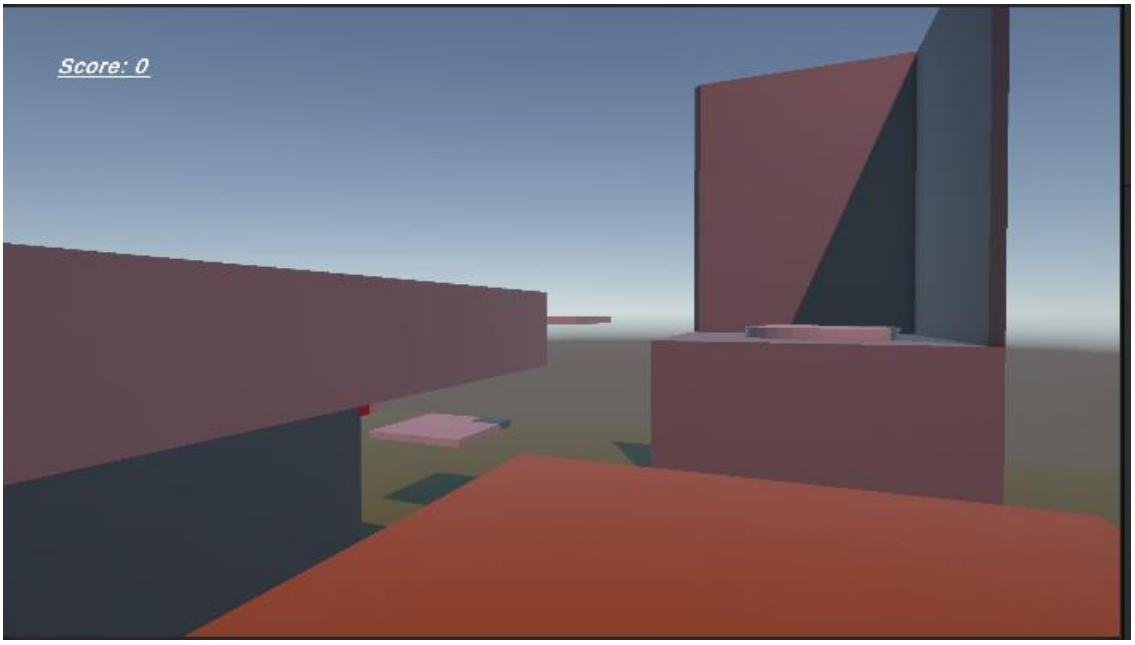
The enemies will be designed to suit the areas they appear in. Scrap monsters and more demonic looking creatures in the Stackland wreckage, whereas the caves have enemies like bats and insects. More complex enemy behaviors may be added if time allows, but the priority will be to make sure the existing enemy interactions work correctly.

Character Switching



This is an optional addition, the idea that a second character be modelled and then during the final stage, you switch to the other character, like above a small dialogue box will to show you are being changed to another character with text explaining an in game reason why you would need to be switching characters. This is a feature which I hope will look more technical than it really is, to help make the game stand out. Inspiration for this idea was taken from Sonic the Hedgehog from 2006,

UI



The UI will be simple, with a legible font, and a good size that the player can see easily. Additional features such as a 2D sprite of Hank made in Aseprite will be made and be set next to a life bar. Below is an example of inspiration

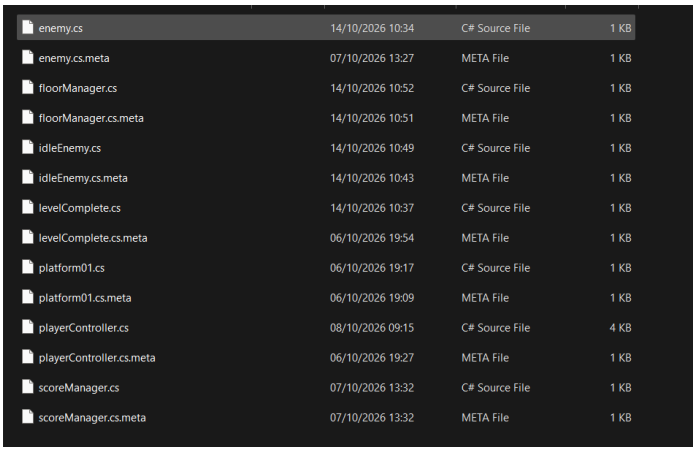


I may potentially try import my own font, but this is will only be done if there is time left over.

Technical Assets

Hank's Odyssey is being programmed in C# and built within the Unity engine. The code is being written in Visual Studio Code and the project is being hosted on Github. The original plan was to use UPBGE (Blender Game Engine) and be programmed in Python, however

this was switched after UPBGE proved to be difficult and unreliable. Unity is installed both on my personal laptop and the college computer where I am stationed.



enemy.cs	14/10/2026 10:34	C# Source File	1 KB
enemy.cs.meta	07/10/2026 13:27	META File	1 KB
floorManager.cs	14/10/2026 10:52	C# Source File	1 KB
floorManager.cs.meta	14/10/2026 10:51	META File	1 KB
idleEnemy.cs	14/10/2026 10:49	C# Source File	1 KB
idleEnemy.cs.meta	14/10/2026 10:43	META File	1 KB
levelComplete.cs	14/10/2026 10:37	C# Source File	1 KB
levelComplete.cs.meta	06/10/2026 19:54	META File	1 KB
platform01.cs	06/10/2026 19:17	C# Source File	1 KB
platform01.cs.meta	06/10/2026 19:09	META File	1 KB
playerController.cs	08/10/2026 09:15	C# Source File	4 KB
playerController.cs.meta	06/10/2026 19:27	META File	1 KB
scoreManager.cs	07/10/2026 13:32	C# Source File	1 KB
scoreManager.cs.meta	07/10/2026 13:32	META File	1 KB

Blender/Maya

The game will contain numerous 3D assets, the environments I plan to have them very detailed and filled with models with detailed textures and make sense structurally, there is no need for a zoo in this game for example. The models will be created in both Blender and Maya, I am familiar with both software, I plan to have environmental pieces created in blender and textured in either Blender or an external software like Photoshop. Maya I plan to use for characters, there will not be many so Maya will not be used as often. All models will be exported as FBX so they can be used within Unity, however I will keep the .blend and .mb files safe, in case I want to make changes, if they get lost, there is the ability to import the FBX files back into the 3D modelling software and edit them that way.

2D Illustrations

The 2D illustrations will be created using Adobe Photoshop and Sketchbook Pro. These will act as the basis for the cutscenes in the game, since animated 3D models will not be possible in the time given to complete the FYP.

Audio and Sound Effects

Sound effects will be used to make the game feel more responsive. For example, a metal crunch sound will play when an enemy is defeated, or a jump sound effect. Voice acting is also a possibility for dialogue or cutscenes, this will be done to make the game feel more alive, stock sound effects may also be used to not infringe on any copyrights, Audacity will be used for recording audacity, which I used for an animatic assignment in 3rd year.

16:9 Storyboards Title Storyboard 02 By Liam Treacy Date 12-11-25


Establishing shot: Computer Lab is working hard, however there is an eerie atmosphere in the air.


Mr Black enters, Arthur (our protagonist) is worried, a dark shadow emits when Mr Black enters.


Mr Black looks down, he looks irritated, he is getting his assignments ready to hand out.


Mr Black has a twisted look on his face, he hands Arthur his assignment, Arthur is like a deer in headlights.


"Arthur, my favorite. This assignment will be your finest work, no doubt." Arthur hides behind screen.


Mr Black hands Arthur the assignment, behind the computer, faces intentionally hidden, background very dark

16:9 Storyboards Title Storyboard 03 By Liam Treacy Date 12-11-25


Arthur looks at his screen He is nervous, worry overwhelms him


Cut to a blank computer screen


Arthur decides to type something on his keyboard, the look of determination is inspiring


The comment code is typed, huge progress is being made


Close up of Arthur, he is nervous, some awe inspiring italian music plays.


The clock strikes 1pm, class is over, Arthur spent too long in his fantasy land.

16:9 Storyboards Title Storyboard 01 By Liam Treacy Date 12-11-25


Arthur looks down at the brief, his desk is rather messy, he reads the Hello World requirement.


Internally ("What is this?, he cannot be serious?"), arthur is staring down at the assignment.


Arthur turns, and looks up to Mr Black. Arthur is kinda pissed.


Mr Black, cannot contain his pleasure, he holds a smug, evil grin on his face


A small fantasy occurs, where Mr Black transforms into the Devil, colors become very dark reds and oranges.

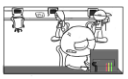

"Good luck, (whisper) "loolooo", Arthur cannot contain his distaste for Mr Black.

16:9 Storyboards Title Storyboard 05 By Liam Treacy Date 12-11-25

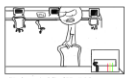

"Aww Shit" Arthur looks back to the clock, he sees he is out of time. He looks "God"


"Well, well, well, Class is over." Mr. Black looks delighted with Arthur's Dilemma.

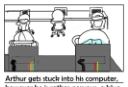

"EVERYBODY GET OUT!"


Arthur looks nervously at his computer, he sees what little work he has done, he really wasted the class


Final cut to screen, shown he wrote 2 comments, save icon appears in bottom left


"You know what? ChatGPT will do" Arthur got up to classic arsonist she is pissed, he has accepted his circumstances, dark shading disappears

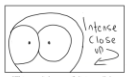
16:9 Storyboards Title Storyboard 03 By Liam Treacy Date 12-11-25


Arthur gets stuck into his computer, however he is rather nervous, a blue glow emits from the screen.


Arthur looks at his screen, a look of determination on his face.


The look of confidence changes to a rather puzzled expression, background blurred to focus on Art.


"Alright, this won't be so bad, it's just one function, this will be easy." Arthur gets comfy in his chair.



"This won't be easy" (Internally) Reference to Homer Simpson and his frequent internal monologues


Reality hits Arthur, he looks at his computer and realises he needs to get a move on.

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